OmniStudy: Local LLM AI Academic Study Assistant

Student: Rhydham Shukla | **Course:** Machine Learning August 2026 Batch, Unlox Academy | **Minor Project:** 5 (Ollama + Streamlit)

System Architecture & Hardware Specs

User Interface:	**Streamlit Web Framework**
Local Inference Engine:	**Ollama REST API (localhost:11434)**
Hardware Accelerator:	**NVIDIA GeForce RTX 5050 Laptop GPU**
Active Foundation Model:	**deepseek-r1:7b (Q4 Quantized)**
Average Benchmark Latency:	**21.31 seconds**

Core Implementation Highlights

- **REST Integration:** Built requests wrapper communicating directly with Ollama local generate endpoints without cloud dependencies.
- **Chain of Thought Parser:** Implemented regex parsing to separate raw thinking tokens inside think tags into an interactive UI expander.
- **Model Switching:** Automated API health check to query available models and prioritize deepseek-r1:7b for academic math.

APPLICATION INTERFACE & VERIFIED LIVE GPU INFERENCE

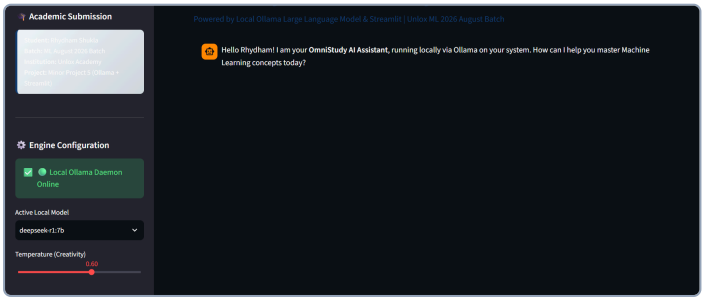


Figure 1: OmniStudy Dashboard with Curriculum Prompts and Settings

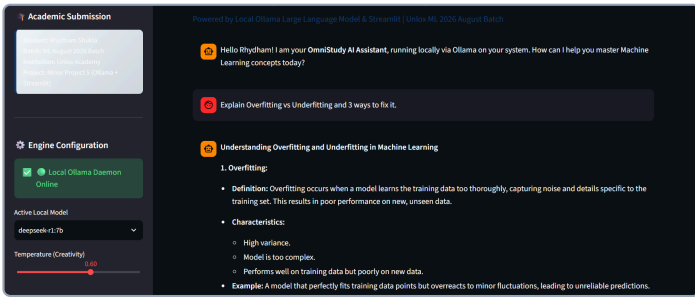


Figure 2: Verified Completed Answer with Reasoning Expander and Latency Badge

EMPIRICAL 5 PROMPT BENCHMARK RESULTS (DEEPSEEK-R1:7B ON RTX 5050)

#	Curriculum Topic	Latency	Verified Model Explanation Summary
1	Overfitting vs Underfitting	**19.73s**	Overfitting captures noise (high variance), underfitting fails to capture pattern (high bias). Remediated by regularization, cross validation, and dropout.
2	Sigmoid in Logistic Regression	**25.14s**	Constrains outputs strictly between 0 and 1, naturally representing valid probabilities and pairing cleanly with log loss optimization.
3	Random Forest vs Decision Trees	**19.11s**	Bagging and feature subsampling decorrelate individual trees, reducing variance and preventing single tree overfitting.
4	Stratified Splits for Imbalance	**17.65s**	Standard random split can leave test sets with zero minority cases. Stratified sampling preserves the exact 90/10 ratio across train and test sets.
5	Outlier Capping in OLS	**24.9s**	Because OLS squares errors, extreme listings ($10,000) drag the regression line. 95th percentile capping stabilizes coefficients for typical listings.